FlyRank AI Internship — Week 2 Deliverable

Assignment: The Prompt Ladder (5 Iterations + Baseline)

Track:	General AI Fluency
Phase:	Foundations (Week 2)
Workload:	2 Hours
Intern:	Abdul Raheem
Status:	Ready for Submission

Executive Summary & Voice Card Banner

This deliverable demonstrates disciplined prompt engineering through a systematic 5-step ladder process (plus 1 weak baseline prompt). Every iteration adds exactly one named layer to solve a specific defect in the output. The evaluation tracks empirical improvements in output quality, identifies honest failure modes (including a prompt change that degraded output quality), and concludes with a production-ready, stranger-reusable prompt template.

■■ **Voice Card (6 words):** "direct, technical, precise, metrics-driven, zero fluff"
Rule: All project framing, copy generation, and case study beats must eliminate corporate buzzwords ("results-driven", "cutting-edge", "innovative", "seamless") and prioritize verifiable empirical data (F1 score, latency ms, benchmark links).

1. Run 0: Weak Baseline Prompt

```
Help me write portfolio copy for my machine learning project.
```

Baseline Output Excerpt:
"I am a passionate and results-driven AI developer who created an innovative machine learning project. This cutting-edge application leverages state-of-the-art algorithms to analyze data seamlessly and deliver high-impact results for users..."

- **What Changed in Prompt:** N/A (Original weak baseline prompt).
- **What Improved in Output:** N/A (Establishes baseline). Output is vague, full of corporate marketing fluff, and contains zero technical data.
- **What Still Failed:** Embarrassingly generic; lacks domain specifics, engineering decisions, metrics, or structure.
- **What I Would Try Next:** Introduce a clear structural goal by mandating a 3-beat case study format (Problem, Key Decisions, Outcomes).

2. Run 1: Version 1 — Layer Added: Clearer Goal

```
Help me write portfolio copy for my machine learning project.
Structure it specifically as a technical 3-beat case study consisting of:
1. Beat 1: The Problem
2. Beat 2: What I Did & Key Decisions
3. Beat 3: What Came Of It (Verifiable Outcomes)
```

Output Excerpt (V1):
Beat 1: The Problem: Machine learning projects are often difficult for users to evaluate because they take too long to run...
Beat 2: What I Did & Key Decisions: I trained a custom vision model and developed a web front-end...
Beat 3: What Came Of It: The model ran faster, had higher accuracy...

- **What Changed in Prompt:** Added explicit structural requirements (3-beat case study framework).

- **What Improved in Output:** Transformed from a single rambling paragraph into three distinct, structured engineering sections.
- **What Still Failed:** Relies entirely on generic placeholders ("custom vision model", "ran faster") with zero concrete metrics.
- **What I Would Try Next:** Supply real context containing specific frameworks, baseline latency, precision metrics, and dataset sizes.

3. Run 2: Version 2 — Layer Added: Real Context

```
Help me write portfolio copy for my machine learning project.
Structure it as a 3-beat case study (Problem, Key Decisions, Outcomes).
Context: PyTorch vision model on FP16 mixed precision. Baseline: 5-sec boot Jupyter notebook. Solution: CPU l
atency 12ms/frame, 0.94 F1 score on 10,000 validation frames. Web UI: Vanilla JS/HTML <100KB payload.
```

Output Excerpt (V2):
"I converted precision from standard FP32 (32-bit floating point) to FP16 mixed precision (16-bit floating point). FP16 is a quantization technique that reduces memory bandwidth by half... Then I built a web page using HTML and Vanilla JavaScript without external libraries like React or Vue..."

- **What Changed in Prompt:** Appended empirical technical context (PyTorch FP16, 12ms CPU latency, 0.94 F1, 10k frames, <100KB UI).
- **What Improved in Output:** Output incorporates concrete technical facts and verifiable metrics (12ms, 0.94 F1, 100KB payload).
- **What Still Failed (■■ Honest 'This Made It Worse' Moment):** The LLM patronized the reader by defining elementary concepts ("FP16 is 16-bit floating point...", "React or Vue are external libraries..."), inflating word count and turning a case study into an introductory tutorial.
- **What I Would Try Next:** Define a specific technical audience so the model assumes high technical literacy and stops defining basic terms.

4. Run 3: Version 3 — Layer Added: Defined Audience

```
Target Audience: Senior Machine Learning Hiring Manager or Staff AI Engineer evaluating technical candidate c
apabilities.
```

Output Excerpt (V3):
"To eliminate overhead, I trained an end-to-end PyTorch vision model utilizing FP16 mixed precision, balancing memory footprint with precision. I replaced heavy web frameworks with a lightweight Vanilla JS/HTML interface styled with raw CSS variables..."

- **What Changed in Prompt:** Defined target persona as Senior ML Hiring Manager / Staff AI Engineer.
- **What Improved in Output:** Completely stopped defining basic terms (FP16, CPU, JS libraries), shifting tone to peer-level engineering communication.
- **What Still Failed:** Reintroduced corporate fluff ("end-to-end vision model", "seamless interactive user experience") and used dense paragraphs instead of bullet points.
- **What I Would Try Next:** Enforce negative constraints banning buzzwords and mandating bullet points.

5. Run 4: Version 4 — Layer Added: Constraints

```
Constraints:
- Enforce 6-word Voice Card: "direct, technical, precise, metrics-driven, zero fluff".
- Banned Buzzwords: "seamless", "innovative", "cutting-edge", "end-to-end", "impactful", "leveraging".
- Format: Beat 2 must be 3 bullet points. Keep Beat 1 under 40 words.
```

Output Excerpt (V4):
Beat 1: Standard ML portfolios stop at Jupyter notebooks...
Beat 2: • Trained PyTorch vision model using FP16... • Replaced web frameworks with Vanilla JS UI <100KB... • Published error distribution analyses in notebooks.
Beat 3: • Achieved F1 score of 0.94 on 10,000 frames...

- **What Changed in Prompt:** Enforced Voice Card, banned corporate buzzwords, limited word count, mandated bullet formatting.
- **What Improved in Output:** Corporate fluff totally purged. Sentences became active, technical, and scannable.
- **What Still Failed:** Beat 3 claims empirical metrics (0.94 F1, 12ms latency), but does not reference specific repository artifact files.
- **What I Would Try Next:** Add verification requirements mandating that all outcome claims link directly to verifiable repository artifacts.

6. Run 5: Version 5 — Layer Added: Verification Requirements

```
Verification Requirements:
- Every outcome metric in Beat 3 MUST be explicitly paired with a verifiable repository artifact link (e.g.,
[Model Notebook w05_model.ipynb], [Baseline Notebook w04_baseline_score.ipynb], [Live Demo UI index.html]).
```

Output Excerpt (V5 - Final):
Beat 3: What Came Of It (Verifiable Outcomes):
• Achieved **F1 score of 0.94** on 10,000 validation frames [Model Notebook w05_model.ipynb].
• Reduced average CPU inference latency to **12ms per frame** [Baseline Notebook w04_baseline_score.ipynb].
• Delivered interactive visual demo under 100KB payload [Live Demo UI index.html].

- **What Changed in Prompt:** Added verification requirements mandating direct artifact links for all outcome claims.
- **What Improved in Output:** Performance metrics are anchored to explicit, verifiable repository artifacts, converting resume claims into proof-backed evidence.
- **What Still Failed:** None. Meets all quality criteria, adheres to Voice Card, targets exact persona, and delivers zero-fluff proof.
- **What I Would Try Next:** Clean up and parameterize into a reusable prompt template.

7. Comparative Output Analysis Matrix

Version	Layer Added	Quality Score	Primary Output Defect	Key Output Improvement
Baseline	None (Original)	1 / 10	100% corporate fluff; zero metrics.	Established weak starting point.
V1	1. Clearer Goal	3 / 10	Generic placeholders ("ran faster").	Organized into 3 logical beats.
V2	2. Real Context	4 / 10	■■ Degraded: Patronized reader defining CS terms.	Cites real numbers (12ms, 0.94 F1).
V3	3. Defined Audience	6 / 10	Soft corporate adjectives & dense text.	Targeted peer engineer; eliminated basic definitions.
V4	4. Constraints	8 / 10	Metrics lacked verifiable repo links.	Purged all fluff; bulleted layout.

V5	5. Verification	10 / 10	None. Production-ready.	Paired outcome claims with verifiable artifact links.

8. Final Reusable Prompt Template

```
# Role & Goal
You are an expert AI/ML Technical Portfolio Editor. Draft a 3-beat technical case study for an engineering pr
oject based on the input data provided below.

# Input Project Parameters
- Project Name: [Insert Project Name]
- Core Tech Stack: [e.g., PyTorch FP16, Vanilla JS, FastAPI]
- Baseline Problem & Friction: [e.g., 5-second boot time notebook]
- Key Technical Decisions: [Insert 2-3 key architecture decisions]
- Verifiable Metrics: [e.g., 12ms CPU latency, 0.94 F1 score]
- Proof Artifact Links: [e.g., w05_model.ipynb, w04_baseline_score.ipynb]

# Target Audience
Senior Machine Learning Hiring Manager or Staff AI Engineer.

# Constraints (Voice Card)
- Enforce Voice Card: "direct, technical, precise, metrics-driven, zero fluff".
- Banned Words: "seamless", "innovative", "cutting-edge", "end-to-end", "impactful", "leveraging", "results-d
riven".
- Do NOT define basic technical concepts. Assume senior engineering literacy.

# Verification Rule
- Every metric in Beat 3 MUST be explicitly paired with a bracketed proof artifact link from Input Parameters
.
```

9. Pass / Revise Audit Checklist

[X] **Six Runs Total:** Baseline plus 5 versions (V1 through V5).

[X] **Single Named Layer Per Version:** Goal, Context, Audience, Constraints, Verification.

[X] **Notes Describe Output Changes:** Notes evaluate empirical output changes rather than prompt edits.

[X] **Honest 'This Made It Worse' Moment:** Documented in V2 (LLM defined basic CS terms).

[X] **Stranger-Reusable Final Prompt:** Parameterized template ready for independent execution.